Lesson 1.8 — Composition of Functions

Big idea: (f ■ g)(x) = f(g(x)) means "feed x into g, then feed g(x) into f." Inside-out, every time. The order matters.

Quick review

Notation: (f ■ g)(x) and f(g(x)) mean the same thing.
How to compute: evaluate g first, then plug that result into f.
Order matters: in general, $f(g(x)) \neq g(f(x))$.
Domain trap: the domain of f ■ g is the set of x in the domain of g such that g(x) is in the domain of f.

Worked example

Worked example. Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find (f ■ g)(3) and (g ■ f)(3).

(f ■ g)(3) = f(g(3)) = f(9) = 2(9) + 1 = 19.
(g ■ f)(3) = g(f(3)) = g(7) = 49. Different answers, same inputs — order matters.

Warm-ups

1. Let $f(x) = x + 5$ and $g(x) = 3x$. Find f(g(2)).
2. Let $f(x) = x^2$ and $g(x) = x – 1$. Find g(f(2)).
3. Let $f(x) = \sqrt{x}$ and $g(x) = x + 9$. Find f(g(7)).

Practice

1. Let $f(x) = 2x – 3$ and $g(x) = x^2$. Find (f ■ g)(x) and (g ■ f)(x), simplified.
2. Let $f(x) = 1/x$ and $g(x) = x – 2$. Find the domain of (f ■ g)(x).
3. Decompose $h(x) = (3x + 1)^5$ into f(g(x)) where g is a polynomial and f is a power function.

Challenge

1. Show that if $g(x) = f^{-1}(x)$, then (f ■ g)(x) = x and (g ■ f)(x) = x. Use $f(x) = 4x – 7$ to demonstrate.
2. Sales tax is 8% and a coupon takes $5 off. Let T(x) = 1.08x (tax) and C(x) = x – 5 (coupon). On a $50 item, compare the total cost of (T ■ C)(50) (coupon first, then tax) vs. (C ■ T)(50) (tax first, then coupon). Which is cheaper, and by how much?

Answer key — Lesson 1.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $f(g(2)) = f(6) = 11$.
2. $g(f(2)) = g(4) = 3$.
3. $f(g(7)) = f(16) = 4$.

Practice

1. (f ■ g)(x) = $2x^2 - 3$. (g ■ f)(x) = $(2x - 3)^2 = 4x^2 - 12x + 9$.
2. Need $g(x) \neq 0$ (so f is defined): $x - 2 \neq 0 \Rightarrow x \neq 2$. Domain: $(-\infty, 2) \cup (2, \infty)$.
3. $g(x) = 3x + 1$, $f(x) = x^5$. Then $h(x) = f(g(x))$.

Challenge

1. $f^{-1}(x) = (x + 7)/4$. $f(f^{-1}(x)) = 4 \cdot (x + 7)/4 - 7 = x$. $f^{-1}(f(x)) = (4x - 7 + 7)/4 = x$. ✓
2. (T ■ C)(50) = T(45) = \$48.60. (C ■ T)(50) = C(54) = \$49.00. Coupon-first is cheaper by \$0.40 (the tax on the \$5 you didn't pay).